AN ANALYSIS OF CURRENT TECHNOLOGY ADOPTION, DEMOGRAPHICS, AND FUTURE TRENDS

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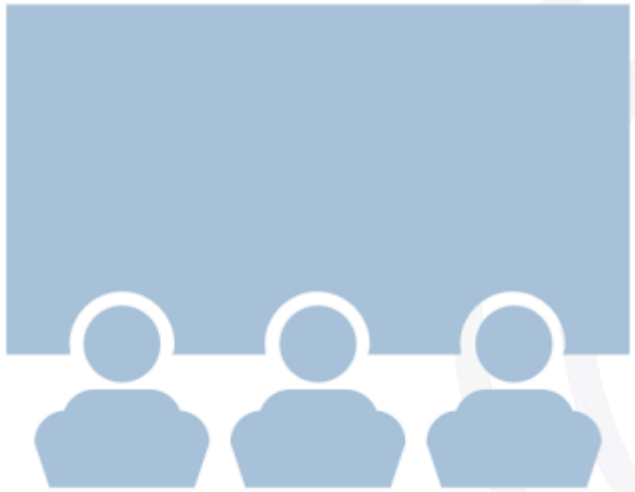
OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
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- Conclusion
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EXECUTIVE SUMMARY

Introduction:

In this project report, we aim to provide a detail analysis of the current state of technology adoption, future technology trends, and demographic insights within the programming industry. Our investigation covers a wide range of topics, including the most popular programming languages, databases use today. We also explore the emerging technologies that professionals are eager to learn. By examining these we have come with some interesting findings

Current Technology usage report

- Top 10 Programming Languages :
  - JavaScript,
  - HTML/CSS,
  - SQL and 7 more
- Top 10 Databases worked with :
  - MySQL,
  - Microsoft SQL Server,
  - PostgreSQL and 7 more
- Top 10 Platforms used:
  - Linux,
  - Windows,
  - Docker and 7 more
- Top 10 web frameworks among developers:
  - jQuery,
  - React.js
  - Angular/Angular.js and 7 more

EXECUTIVE SUMMARY

Future Technology trend Report

- Top Programming languages desired next year:
  - HTML/CSS,
  - JavaScript,
  - C# and 7 more Top
- 10 Databases desire for next year:
  - PostgreSQL,
  - MongoDB,
  - Redis and 7 more
- Next years most desired platforms:
  - Linux,
  - Docker,
  - AWS and 7 more

Demographics Report:

- Gender distribution on genders:
  - Men
  - Women
- Respondent count by country:
  - USA,
  - India,
  - UK and so on
- Respondent count by age:

  Most of them were young 28 to 38 with highest responses.

INTRODUCTION

Current Technology usage report (Intro)

- This report gives a detailed analysis of current technology usage among developers, including the most popular programming languages, databases, platforms, and web frameworks.
- I have analyzed data based on a wide range of respondents to identify the top technologies that are shaping the industry of programming today.

Future Technology trend Report

- This report covers the future technology trends which developers expect to see and the programming languages, databases, and platforms they want to use in the coming year.
- Based on the analysis of the survey, I have provided the top technologies for which developers would want to learn and add to their work process, hence giving an outline of what is to come for the tech industry.

Demographics Report:

- The following report details complete demographic analysis within the tech industry on gender distribution, country-wise respondent count, and

METHODOLOGY

Current Technology Usage Report:

- Analysis Method: Cluster analysis was used to identify clusters or segments of developers, according to their current technology usage.
- Data Analyzed: For the analysis, data related to the top 10 most used programming languages, top 10 databases worked with, top 10 platforms used, and top 10 web frameworks have been considered.

Future Technology Trend Report:

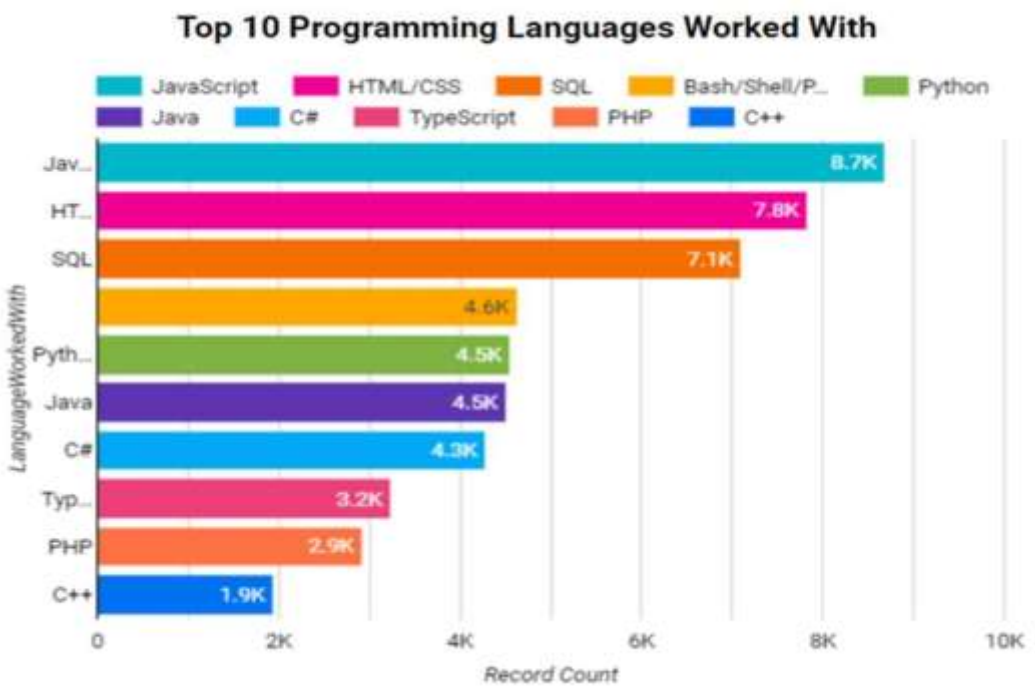
- Analysis Method: Cluster analysis was not utilized. Instead, regression analysis was employed to understand the relationship between different variables and future technology trends.
- Data Analyzed: The analysis focused on predicting future technology trends by examining the top programming languages desired next year, top 10 databases desired for next year, and next year's most desired platforms.

Demographics Report:

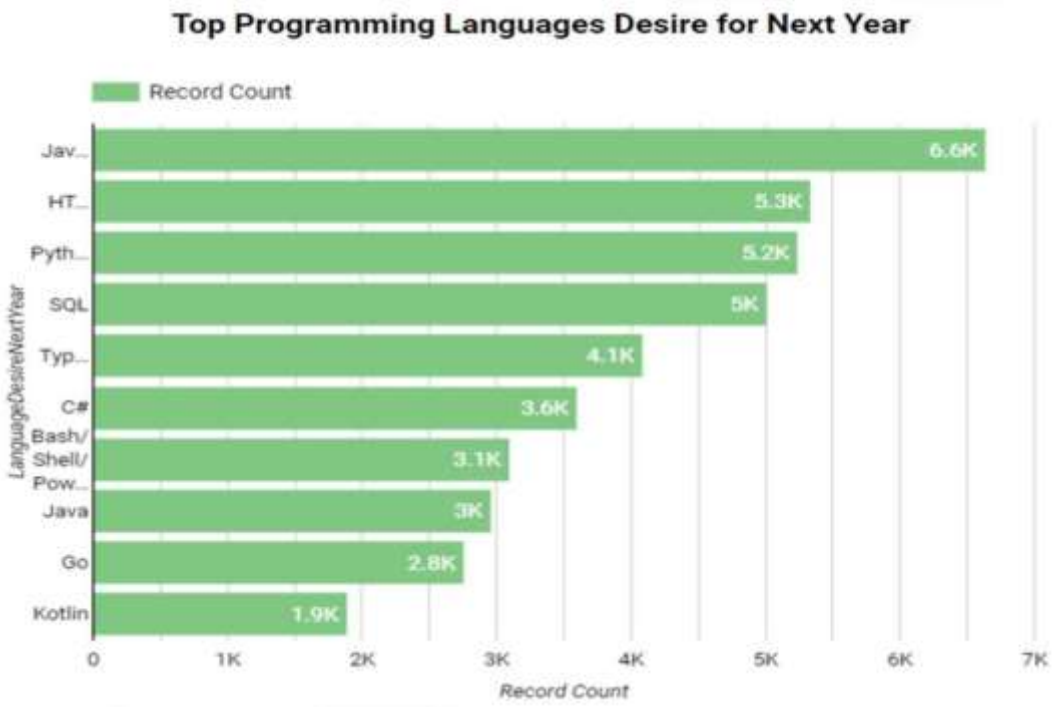
- Analysis Method: Regression analysis was employed to explore the relationship between demographic variables (such as gender, country, and age) and the technology preferences or behaviors of respondents.
- Data Analyzed: The analysis focused on examining gender distribution among respondents, respondent count by country, and respondent count by age to understand the demographic makeup of the surveyed population.

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

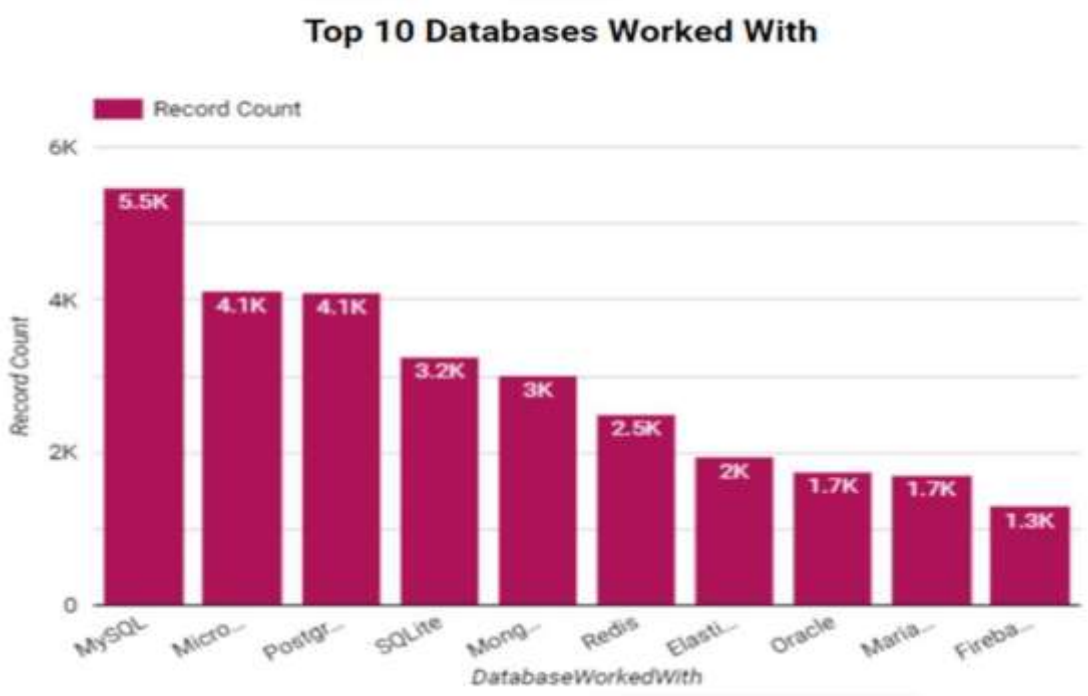
- JavaScript is the most in-demand and desired language. Across both graphs, JavaScript holds the top spot.
- Web development languages are still desired. Both graphs show HTML/CSS, SQL as in-demand languages.
- SQL is the third most popular programming language.
- Developer preferences are shifting. Java goes from third in the 1st graph to sixth in the 2nd graph.
- Conversely, Python goes from seventh in the 1st graph to third (tied) in the 2nd graph

Implications

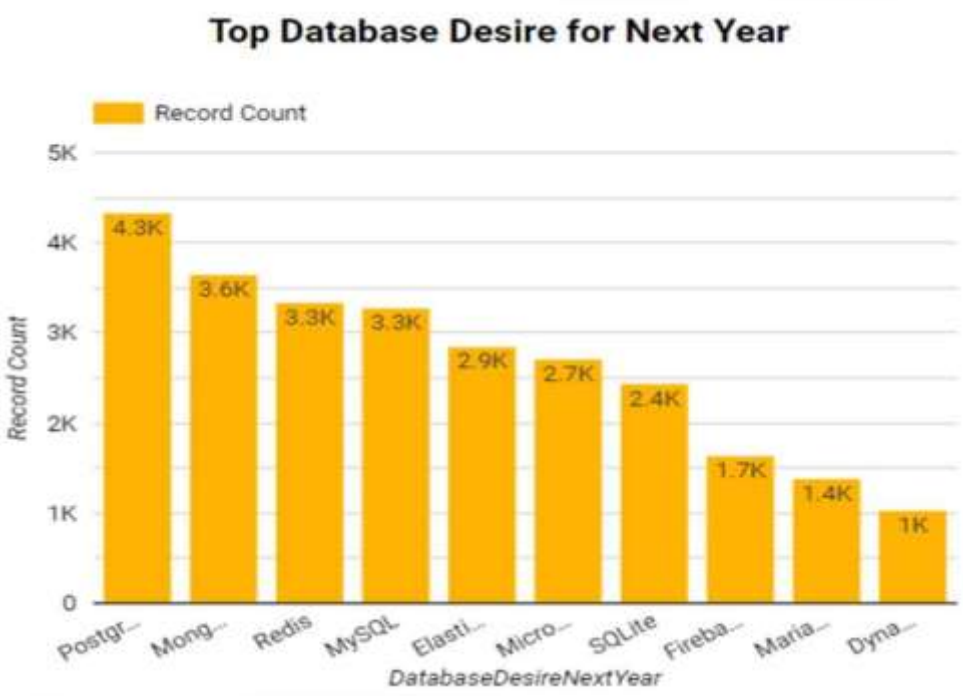
- JavaScript skills are a safe bet for software development careers. JavaScript reigns supreme in both proficiency and desire in-use
- Full-stack developer skills are becoming increasingly important. The continued high demand for HTML/CSS, SQL alongside JavaScript suggests a growing need for developers who can handle both front-end and back-end development.
- The rise of Python reflects a growing data science field. Python's movement up the list suggests an increase in demand for data science skills. As data continues to be a driving force in many industries, this trend is likely to continue.

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- There is a shift in preference from traditional relational databases (MySQL, PostgreSQL) to NoSQL databases (Firebase, MongoDB, Redis).
- While MySQL is currently the most used database, it is not the most desired for next year.
- Firebase, which is not currently being used much, is the most desired database for next year.
- There is a decrease in desire to use currently popular databases like PostgreSQL, MySQL, and SQLite next year.

Implications

- This trend suggests a need for greater scalability and flexibility in data storage
- NoSQL databases may be better suited for handling the increasing complexity of data.
- Organizations may need to invest in training and resources to adopt NoSQL technologies.
- The decision of which type of database to use will depend on the specific needs of the project.

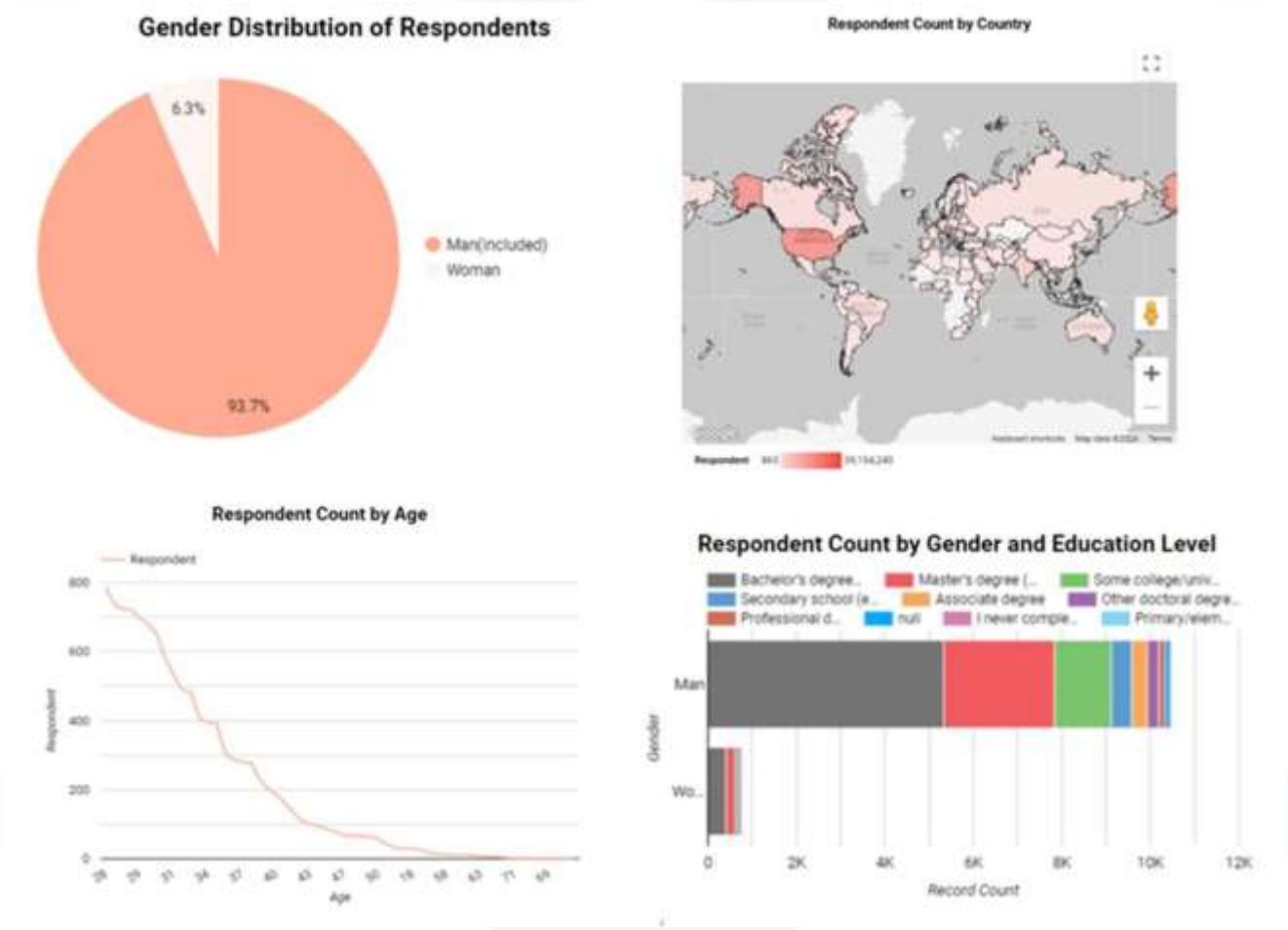
DASHBOARD

https://github.com/KarishmaGoyal/Data-Analyst-Portfolio

DASHBOARD TAB 1

DASHBOARD TAB 2

DASHBOARD TAB 3



DISCUSSION

- JavaScript remains king, but the tide is shifting. While web development languages (HTML/CSS/SQL) hold strong, developer preferences are evolving. Python's rise reflects a growing data science field, suggesting a future with more data-driven roles.
- Overall, the findings suggest a move away from traditional relational databases towards NoSQL databases. This trend is likely being driven by the need for greater scalability and flexibility in data storage.

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript reigns supreme in both charts: the most demanded and desired language for software development.
- Web Development Strong: HTML/CSS and SQL continue to be extremely important for web development consistently ranking in the top few for both current use and desired languages.
- Shifting Preferences: Developer preference is shifting. Traditional languages such as Java and C are still widely used, yet their share is declining. The rise of Python shows interest in the area of data science.

Implications

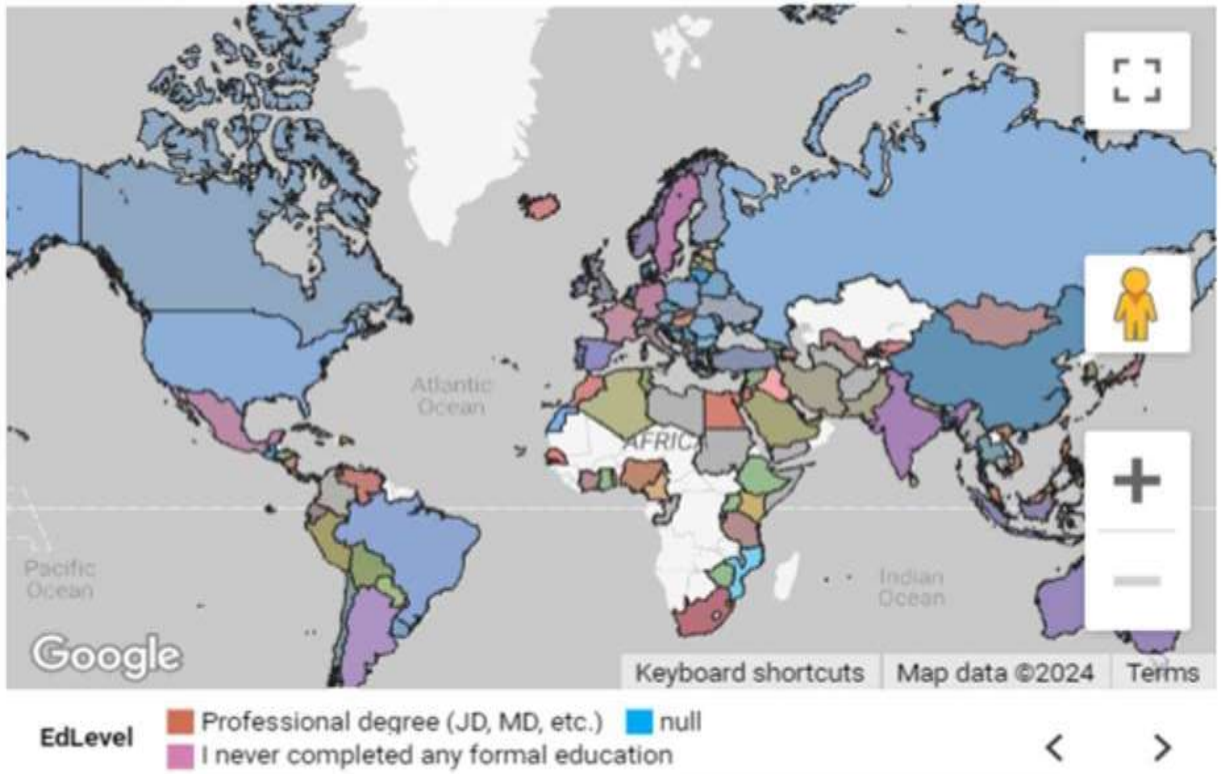
- Invest in JavaScript skills: Being proficient in JavaScript could be an asset, considering it is the most used and most desired language in the industry right now.
- Focus on Full-Stack Development: The high demand for HTML/CSS, SQL coupled with JavaScript reflects the increased need for developers capable of full-stack development.
- Demand for Data Science Skills: With the growing popularity of Python, more and more people need data science skills that are valuable in today's data-driven world.

CONCLUSION

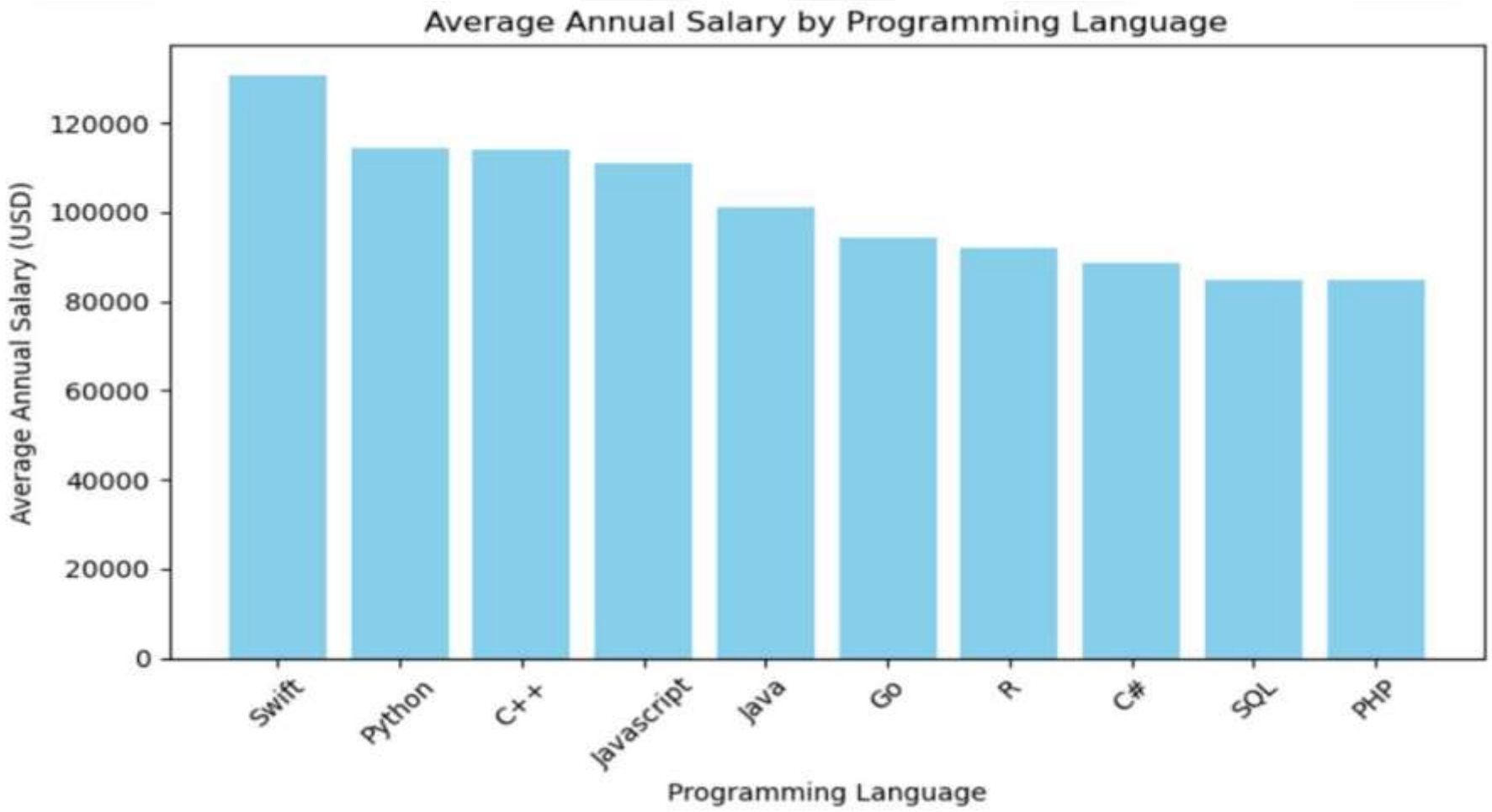
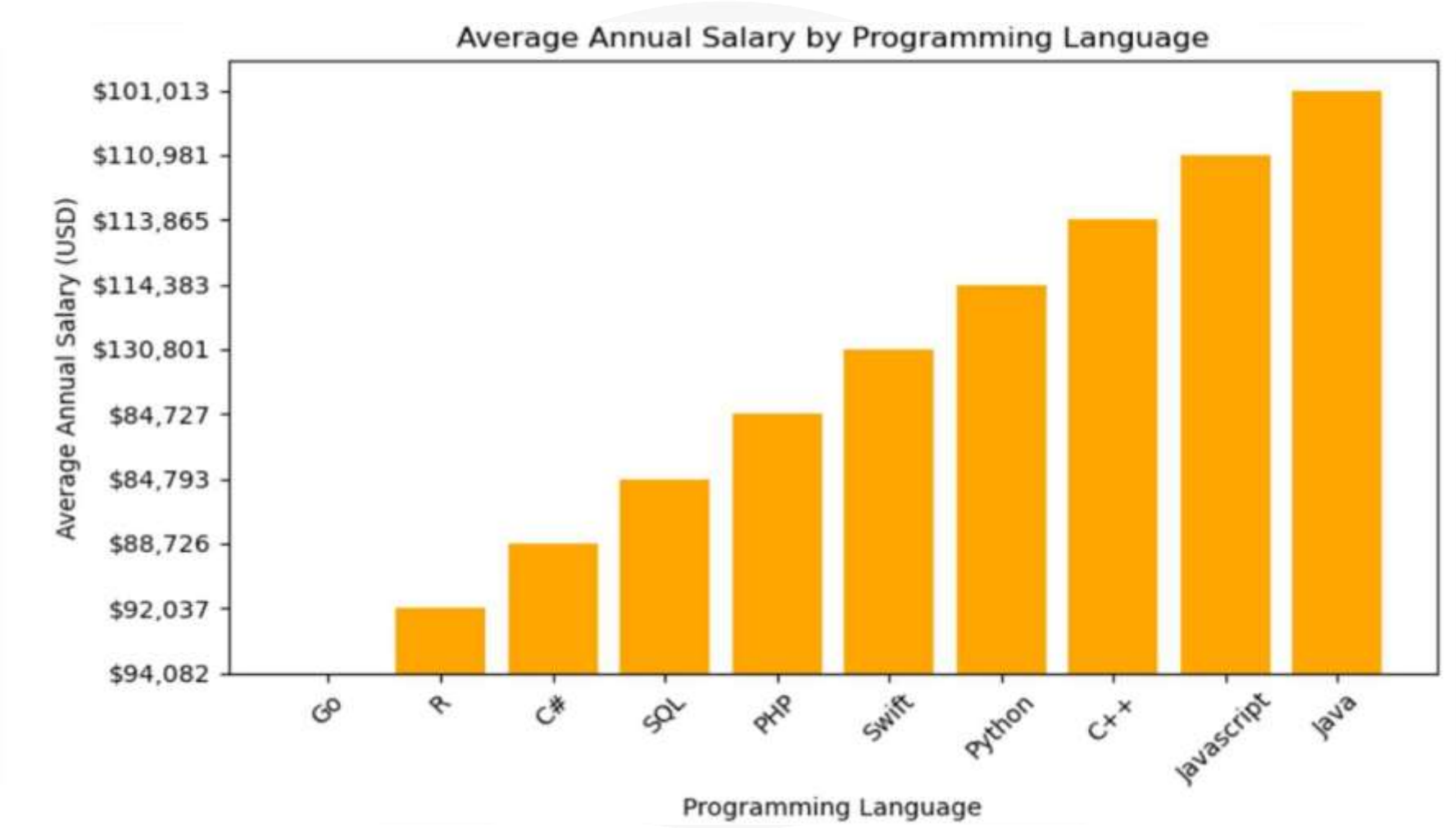
- Developer preferences keep changing. While JavaScript stands atop the table, the uprising of Python and the movements of traditionally prominent languages like Java and C indicate the need to keep flexible and learn new technologies constantly.
- There's a growing demand for developers to bridge front-end and back-end development, coupled with data analysis.
- Popularity with Python reflects the ever-increasing importance of data science. Nowadays, more data is collected and used by businesses than ever, creating a strong demand for developers who can build applications and draw insights from the data.

APPENDIX

Country by Education Level



JOB POSTINGS



POPULAR LANGUAGES